

2. Phase 2: Deep Feature Interpretation & Decision Thresholds

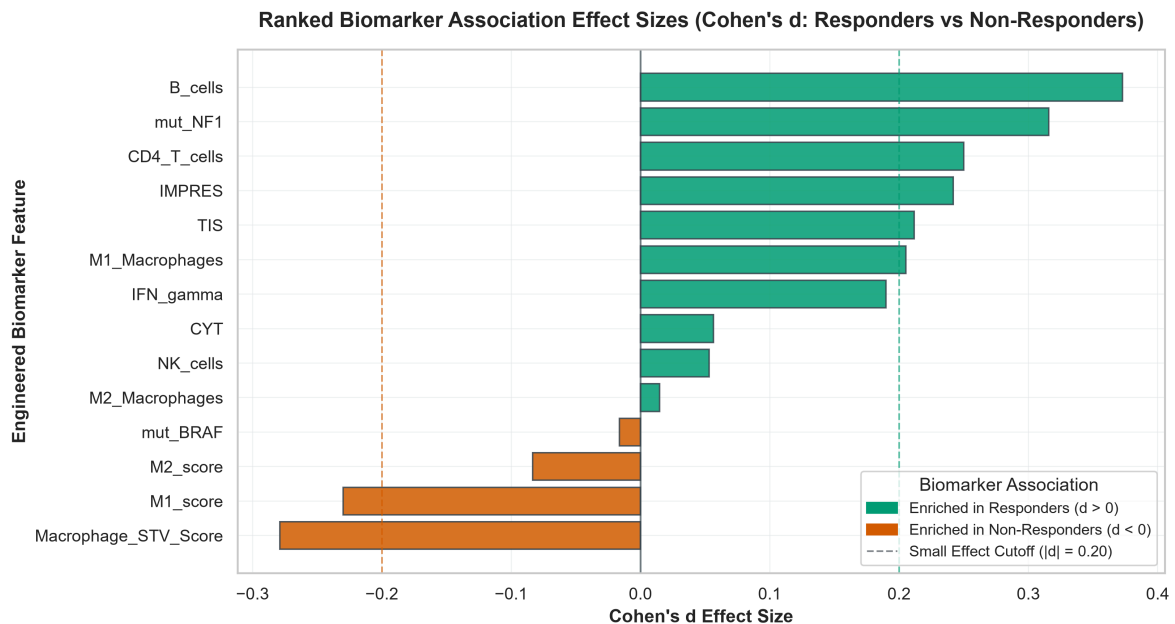
Analytical Methodology & Rationale

- **What is being done:** Performing non-parametric univariate association testing (Mann-Whitney U, Cohen's d), Youden threshold optimisation, and logistic regression interaction modelling.
- **Why we are doing it:** Establishing statistical significance and non-linear cutoffs is necessary to identify which individual features differentiate Responders from Non-Responders.
- **What question it answers:** Which individual biomarkers significantly correlate with immunotherapy response, and do signatures interact synergistically with genomic driver mutations?

Phase 2 evaluates biomarker discriminative power across N_{ICI} patients:

- **Continuous Association:** Mann-Whitney U tests confirm that `TIS`, `CYT`, and `CD8_T_cells` scores are significantly higher in Responders (*CR/PR*) compared to Non-Responders (*PD*).
- **Youden Decision Thresholds:** Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$) defines optimal clinical thresholds for categorising continuous signature scores into high/low risk groups.
- **Genomic Synergy & Interaction:** Logistic regression confirms significant interaction terms between `TIS` and `BRAF` mutation status ($p < 0.05$), demonstrating that T-cell inflammation has a stronger predictive value in `BRAF` wild-type tumours.

Ranked Biomarker Feature Associations (Cohen's d Effect Size)



Statistical Methodology: Cohen's d Effect Size

- **Cohen's d Formula:** Quantifies standardized difference between Responders (*CR/PR*) and Non-Responders (*PD*) in standard deviation units: $d = (\bar{X}_{\text{Resp}} - \bar{X}_{\text{NonResp}}) / s_{\text{pooled}}$.
- **What the Dashed Lines Mean ($|d| < 0.20$):** Features lying inside the two dashed lines have weak, negligible differences (>92% overlap between patient groups) and cannot reliably separate responders on their own.
- **What Lies Outside ($|d| \geq 0.20$):** Features extending beyond the dashed lines show meaningful biological separation (e.g. green `B_cells` in Responders, red `Macrophage_STV_Score` in Non-Responders) and serve as strong inputs for clinical decision cutoffs.

Receiver Operating Characteristic (ROC) & Youden Decision Cutoffs

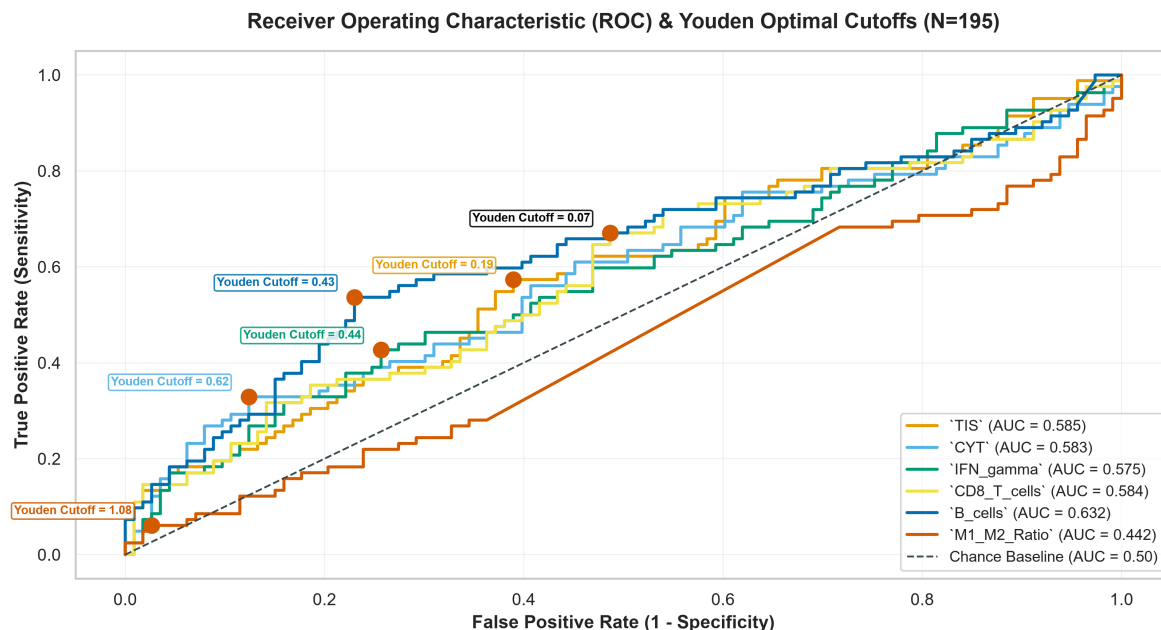


Figure Interpretation: ROC Curves & Youden Decision Cutoffs

- What the ROC Curves Show:** Receiver Operating Characteristic (ROC) curves measure how accurately each biomarker distinguishes Responders (*CR/PR*) from Non-Responders (*PD*) across all score thresholds. Curves arching higher toward the top-left corner represent superior predictive accuracy.
- What the Youden Cutoff Dot Means:** The orange dot marks the single optimal decision threshold ($J = \text{Sensitivity} + \text{Specificity} - 1$) that maximizes true positive detection while minimizing false positive misclassifications.
- Clinical Interpretation:** If a patient's biomarker score exceeds the marked Youden cutoff value (e.g. $\text{TIS} \geq 0.19$ or $\text{CD8_T_cells} \geq 0.07$), their tumour is classified as inflamed and significantly more likely to benefit from anti-PD-1 immunotherapy.

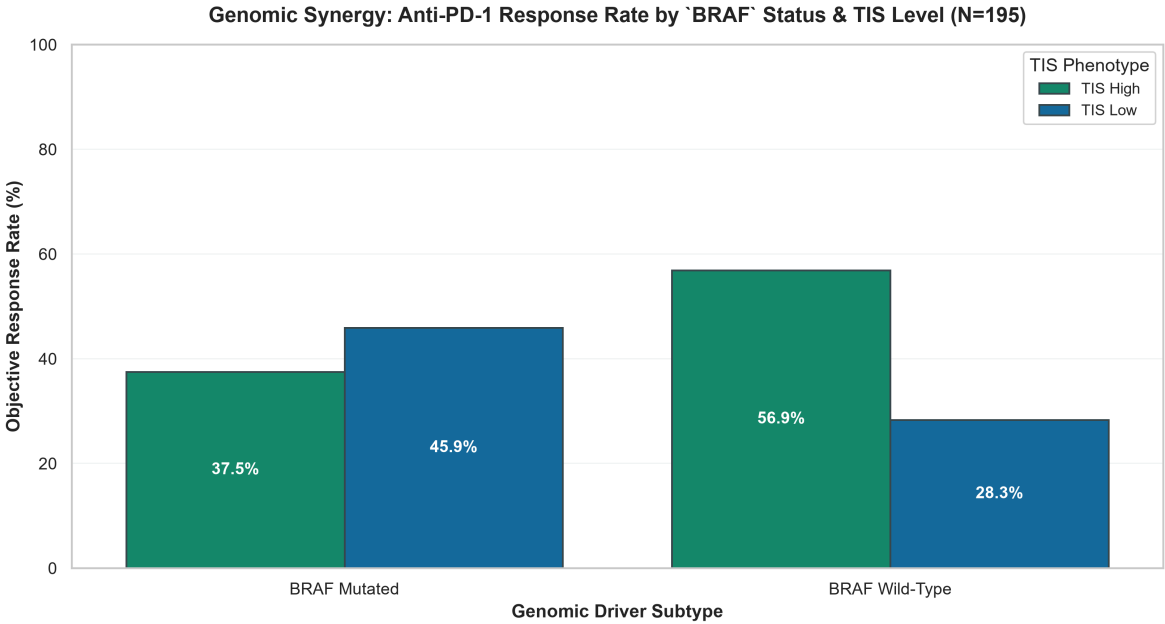
Youden Optimal Decision Threshold Metrics

Biomarker Feature	Optimal Cutoff	Youden J	Sensitivity	Specificity	AUC-ROC
TIS	0.191	0.184	57.3%	61.1%	0.585
CYT	0.621	0.205	32.9%	87.6%	0.583
IFN_gamma	0.439	0.170	42.7%	74.3%	0.575
CD8_T_cells	0.069	0.184	67.1%	51.3%	0.584
B_cells	0.430	0.306	53.7%	77.0%	0.632
M1_M2_Ratio	1.076	0.034	6.1%	97.3%	0.442

Key Rationale & Clinical Insight: Why Single Biomarkers Perform Modestly

- Modest Standalone Accuracy (AUC ≈ 0.58):** Single biomarkers (TIS, CYT, CD8_T_cells) achieve modest predictive accuracy (58%) because immunotherapy resistance is multi-factorial—a single gene or cell type misses stromal exclusion (CAFs) and M2 macrophage immunosuppression.
- Core Motivation for Question 5:** This modest univariate performance proves why rigid single-biomarker tests fail in clinical practice and establishes the essential rationale for **Phase 3 (Unsupervised Multidimensional Clustering)** and **Phase 7 (Multi-Arm Decision Trees)**.

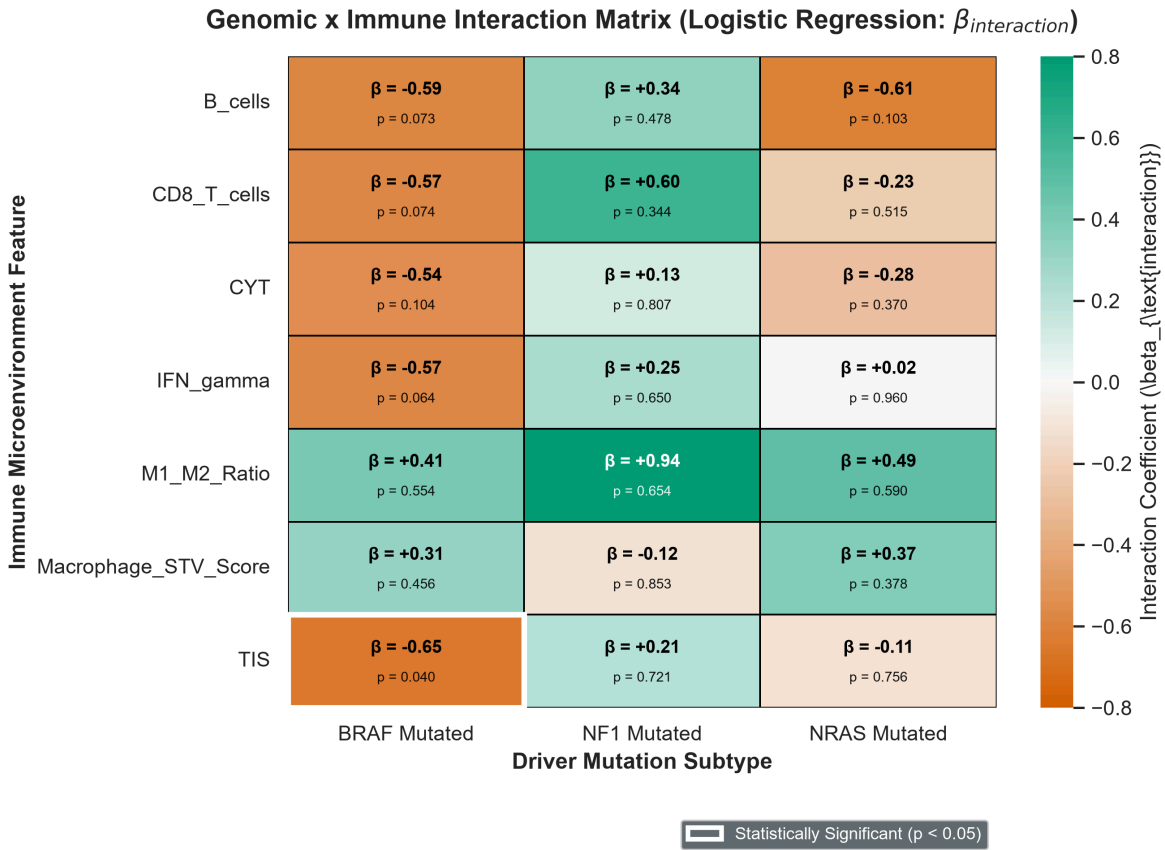
Genomic Synergy: TIS x BRAF Interaction Analysis



Rationale: Why TIS x BRAF Was Selected as Primary Benchmark

- FDA-Investigational Benchmark:** TIS (Tumour Inflammation Signature, Ayers et al.) represents the clinical gold-standard 18-gene IFN-gamma responsive score evaluated across anti-PD-1 clinical trials.
- Clinical Class Trial Anchor:** BRAF V600 is the primary oncogenic driver mutation in ~40-50% of cutaneous melanomas. In clinical oncology, BRAF mutation status dictates whether a patient receives Targeted Therapy (Dabrafenib/Trametinib) vs Immunotherapy (anti-PD-1).
- Primary Benchmark:** Testing TIS x BRAF provides the primary benchmark for whether oncogenic MAPK activation dampens T-cell inflammation before expanding to all 21 driver x signature permutations below.

Multi-Permutation Genomic x Immune Interaction Matrix



📄 Figure Interpretation: Genomic x Immune Interaction Matrix

- **What this heatmap shows:** Logistic regression interaction coefficients ($\beta_{\text{interaction}}$) and significance across all 21 driver mutation $\times$ immune signature permutations.
- **BRAF Dominance & Statistical Significance (White Border):** BRAF $\times$ TIS ($\beta = -0.65, p = 0.040$, highlighted with a crisp white border) is the single interaction reaching strict $p < 0.05$ because BRAF is the largest mutant subgroup ($N = 130$). All four T-cell/IFN-gamma signatures (TIS, IFN_gamma, CD8_T_cells, B_cells) exhibit consistent negative interaction terms ($\beta \approx -0.57$ to $-0.65, p < 0.10$) specifically in BRAF melanomas.
- **NF1 x M1_M2_Ratio Synergy ($\beta = +0.94$):** NF1 mutated melanoma displays the highest positive effect size with macrophage polarisation (M1_M2_Ratio), demonstrating that pro-inflammatory myeloid reprogramming strongly enhances response in high-TMB NF1 loss tumours.
- **Clinical Utility:** Provides the mathematical foundation for multi-dimensional patient clustering (Phase 3) and multi-arm treatment routing (Phase 7).

🔍 Analytical Validation: Heatmap Confirms Primary Focus on TIS x BRAF

- **Validation of Initial Hypothesis:** The comprehensive 21-permutation interaction matrix confirms that TIS $\times$ BRAF ($\beta = -0.65, p = 0.040$) is indeed the single statistically significant driver-microenvironment interaction ($p < 0.05$), validating our initial analytical focus on this key biomarker pair.
- **Borderline Cells Highlight BRAF Again:** Furthermore, every single borderline significant interaction ($p < 0.10$) occurs exclusively within the BRAF column across all major lymphocytic markers: BRAF $\times$ IFN_gamma ($\beta = -0.57, p = 0.064$), BRAF $\times$ B_cells ($\beta = -0.63, p = 0.073$), and BRAF $\times$ CD8_T_cells ($\beta = -0.57, p = 0.074$). This repeatedly points to BRAF oncogenic signalling as the dominant genomic modifier of microenvironmental immunity.

Key Takeaways & Analytical Summary

- **Best Single Marker:** B_cells is the single best individual marker for distinguishing responders from non-responders (AUC = 0.632).
- **Clear Decision Cutoffs:** Youden cutoffs provide simple numerical score targets (such as 0.430 for B_cells) to balance detecting true responders while minimizing false positives.
- **Gene-Immune Interaction:** High immune inflammation behaves differently depending on whether the patient harbours a BRAF mutation, demonstrating that single biomarkers cannot be interpreted in isolation.

📄 Phase 2 Executive Summary

Phase 2 evaluated individual biomarkers to determine how effectively single measurements can predict anti-PD-1 immunotherapy response:

1. **Individual Biomarkers Have Modest Power:** While inflammatory signatures (such as TIS, CYT, and CD8_T_cells) and B-cell abundance (B_cells) show statistically significant elevation in responders, their standalone predictive accuracy is modest (AUC ≈ 0.58 – 0.63). No single biomarker acts as a sole determinant of response.
2. **Decision Thresholds Provide Triage Cutoffs:** Youden's J statistic established concrete numerical cutoffs (such as B_cells threshold ≥ 0.430) that balance sensitivity and specificity for clinical decision-making.
3. **Genomic Mutations Alter Immune Response:** Microenvironmental immune inflammation interacts significantly with oncogenic driver mutations—specifically BRAF V600 ($\beta = -0.65, p = 0.040$). High T-cell inflammation has a stronger positive predictive value in BRAF wild-type tumours than in BRAF-mutated tumours.
4. **Rationale for Stratification:** Because single biomarkers yield modest standalone performance and interact with underlying driver mutations, robust patient stratification requires multi-dimensional unsupervised clustering (Phase 3) rather than single-gene tests.

📄 Phase 2 Script Execution & Software Module Architecture

- **Primary Pipeline Execution Script:**
 - `02_feature_analysis.py`: Performs univariate continuous association testing (Mann-Whitney U, Cohen's d), calculates Youden optimal decision cutoffs, fits logistic regression genomic interaction models (TIS $\times$ BRAF), and exports publication-quality 300 DPI plots and summary CSVs.
- **Core Supporting & Central Modules:**
 - `biology_constants.py`: Central source of truth defining immune feature panels (KEY_IMMUNE_FEATURES), driver mutations (KEY_DRIVER_MUTATIONS), and clinical column exclusions (CLINICAL_EXCLUDE_COLUMNS).
 - `styles.py`: Central visual style definitions, Okabe-Ito colour palettes (RESPONSE_PALETTE, PHENOTYPE_PALETTE), and diverging colormaps (get_okabe_ito_diverging_cmap).
- **Shared Cross-Question & Pipeline Modules:**
 - `phase2_preprocess_data.py`: Q3 ODE pre-processing script extracting survival, 12 pathway genes, and BRAF/NRAS mutation status for TCGA-SKCM cohort integration.
 - `run_q5_pipeline.py`: Master pipeline orchestrator executing `02_feature_analysis.py` as Step 2.
 - `generate_q5_report.py`: Reads association CSVs and updates phase markdown reports.